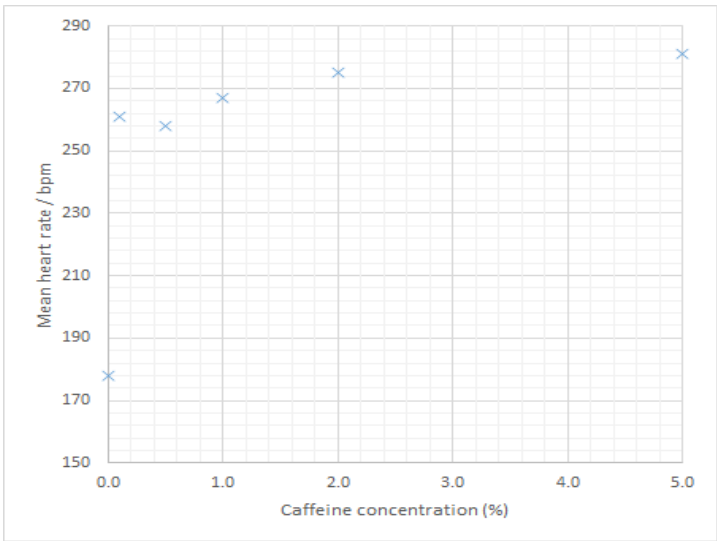


Question number	Answer	Mark
2(a)	• total calculated and divided by 5 Example of calculation: $1405 \div 5 = 281$	(1)

Question number	Answer	Mark
2(b)	Graph plotted to show the following: - labelled axes with correct orientation and linear scale (1) - data plotted as {scatter graph / line graph} (1) - all points plotted correctly (1) Allow ecf from 2a Example graph: 	(3)

Question number	Answer	Mark
2(c)	An answer that includes the following points: - there will be no (significant) correlation (1) - between the caffeine concentration and the (<i>Daphnia</i>) heart rate (1)	(2)

Question number	Answer	Additional guidance	Mark
2(d)	An explanation that includes the following points: • simple nervous system (1) • so less likely to suffer {pain / stress} (1) or • abundant in nature (1) • so not affecting food chain (1)	Accept invertebrate nervous system	(2)

Question number	Answer	Additional guidance	Mark
2(e)(i)	• calculate the value of d^2 (1) • calculate the value of $6\sum d^2$ (1) • calculate the value of r_s (1) Example of calculation: $\sum d^2 = 2$ $6\sum d^2 = 12$ $r_s = 0.943$	Allow ecf from first or second marking point Correct answer with no working shown gains full marks	(3)

Question number	Answer	Mark
2(e)(ii)	An explanation that includes any five of the following points: • as caffeine concentration increases, heart rate increases (1) • critical value is 0.886 (1) • calculated value (0.943) is higher than critical value (1) • therefore reject the null hypothesis (1) • there is a significant positive correlation between concentration of caffeine and heart rate (1) • low concentrations have a large effect, higher concentrations give a smaller increase (1)	(5)

